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# The Algorithm Is Not the Answer: A Methodological Review of Metaheuristic Controller Tuning

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## Abstract

Metaheuristic optimization has become a widely adopted approach for controller parameter tuning in nonlinear, multi-objective engineering systems where classical methods fail. However, the rapid expansion of the literature has not been matched by methodological rigor, raising concerns about reproducibility and practical relevance. This review provides a critical synthesis of studies published between 2025 and 2026, organized thematically around controller architectures, optimizer families, and objective function design. Across five application domains, EV charging, power systems, robotics, process control, and marine systems, we identify persistent gaps: objective functions that disregard engineering priorities, weak constraint handling, unfair benchmarking, poor reproducibility, and insufficient validation. To move the field forward, we offer four prescriptions: benchmark relentlessly, validate physically, attribute honestly, and report completely. We conclude that meaningful progress requires shifting focus from algorithmic novelty to problem-structure-aware, methodologically credible optimization frameworks.

***Index Terms*—Metaheuristic optimization; Controller tuning; Swarm intelligence; Evolutionary algorithms; Systematic review**

## I. Introduction

A key problem in control engineering is the tuning of the controller parameters. The decades of empirical evidence support the fact that classical methods like Ziegler–Nichols, Cohen–Coon, and internal model control work best in linear and low-order systems with well-defined dynamics [1]. Nonetheless, current engineering systems exhibit nonlinear behavior and competing performance objectives that violate the assumptions underlying conventional tuning approaches. Providing gradient-free, global optimization, and non-convexity, multi-modality, and constrained design space, metaheuristic optimization has become an interesting alternative to offer solutions to optimization problems [2]. Consequently, the literature has grown at an accelerated pace and encompasses a wide spectrum of controller structures, optimization schemes, and areas of application. The rapid proliferation of this has not been accompanied by methodological consistency, however. Most studies are based on improvements in performance, which are limited or non-standardized, and frequently lack statistically rigorous comparisons to existing methods. Implicit functions are often limited to simplified criteria of integral errors that do not reflect some fundamental engineering factors like robustness and disturbance rejection. Moreover, validation is often performed in idealized simulation environments, whereby there is a lack of consideration of model uncertainty, parameter variability, and real-time implementation constraints. Such problems make the reproducibility of results difficult and constrain the practical transferability of the methods suggested.

This review overcomes these limitations by critically synthesising original studies published in the period between 2025 and 2026. In contrast with the current reviews, which tend to list algorithms or applications in chronological order, we use a thematic approach that is grouped around three interrelated dimensions: controller families, metaheuristic optimization strategies, and objective function design. This framework enables us to not only discover what has been done, but also what combination works or does not work in particular application cases. We begin by constructing a taxonomic system which systematically classifies the literature in terms of controller, optimizer, and objective dimensions, showing patterns of interrelation that are lost in algorithm-focused reviews. Second, we offer a critical methodological review, which is based on objective function quality, constraint management, fair benchmarking, reproducibility and the depth of validation. Third, we generalize results in key application

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areas, demonstrating that the worth of metaheuristics is highly contingent on the physical and operational properties of each area. Fourth, we develop research gaps and design implications, not just a call to do more research as a generic call but provide actionable recommendations to future research.

## II. Controller Tuning as a Multi-Objective and Multi-Paradigm Optimization Problem

Controller tuning is inherently a multi-objective decision-making problem shaped by competing performance requirements and nontrivial computational structure. As illustrated in Fig. 1, practitioners must balance tracking accuracy, robustness to disturbances and uncertainties, and control effort, which are often in direct conflict. These trade-offs are compounded by two key factors. First, the problem structure is challenging—non-convex, expensive to evaluate, and constrained by implicit stability requirements. Second, the tuning paradigm—offline, online, or hybrid—determines how parameters are optimized and adapted. While metaheuristics address many of these challenges, systematic guidance on paradigm selection remains a critical gap, as explored in the following subsections.

The controller tuning problem is an inherently multi-objective decision-making problem, influenced by conflicting performance needs, and nontrivial computational structure. Practitioners are confronted by the tradeoffs between tracking accuracy, disturbance and uncertainty robustness, and control effort, which are usually mutually conflicting as shown in Fig. 1. These trade-offs are compounded by two key factors. To start with, it has a difficult problem structure, which is non-convex, costly to estimate, and bounded by implicit stability constraints. Second, it is the tuning paradigm, whether offline, online, or hybrid, that is used to determine the parameters that are optimized and adapted. Though metaheuristics can help resolve most of these problems, there is an urgent deficiency in systematic advice on paradigm selection, as discussed in the following subsections.

### A. Problem Structure and Complexity

Controller parameter tuning consists of picking a set of parameters (a vector) $\mathbf{k} \in \mathcal{K} \subseteq \mathbb{R}^n$ that will optimize a performance measure $J(\mathbf{k})$ (under stability and operational limits). A number of properties make this problem different to the typical problems of optimization. First, the objective function $J(\mathbf{k})$ is generally not convex and can have more than one local optima, especially when the controller has nonlinear components or the cost is a combination of contradictory metrics [3]. Second, computation of $J(\mathbf{k})$ needs to compute the closed-loop system, and hence each evaluation of the functions is computationally expensive, and it is impossible to compute the gradient analytically. Third, the feasible set $\mathcal{K}$ is often defined by implicit stability margins, saturation limits, or transient performance bounds that are not easily expressed as simple algebraic constraints [4].

These characteristics justify the shortcomings of classical tuning approaches. Gradient methods have smooth and convex goals and tend to approach local optima. The methods of analysis (e.g. internal model control) presuppose certain plant structures. The heuristic rules (e.g., Ziegler–Nichols) offer sensible initial condition but hardly ever give the best results with uncertainty or competing goals [5]. Metaheuristics overcome these constraints by using gradient-free exploration, global search, and the ability to handle complex objective functions and constraints in a flexible manner.

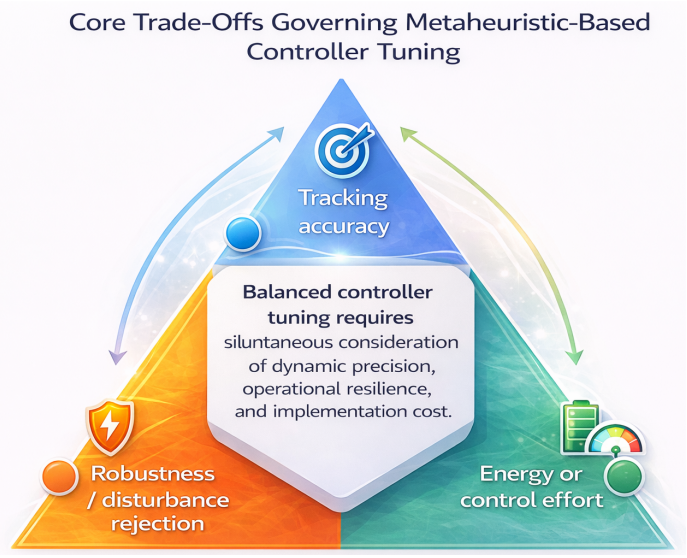



Fig. 1. Core trade-offs in metaheuristic-based controller tuning: balancing tracking accuracy, robustness/disturbance rejection, and control effort.

### B. Offline, Online, and Hybrid Tuning Paradigms

The reviewed corpus shows that there are three distinct tuning paradigms that have varying methodological needs. The most widespread method is offline tuning, which optimizes controller parameters before deployment using a fixed plant model and disturbance scenario [6], [7]. This paradigm is suitable when the dynamics of plants are well-characterized, and operating conditions are predictable, but unmodeled variations or degradation cannot be accommodated.

Online tuning uses constant parameters update in operating conditions, allowing them to adapt to varying conditions. This paradigm is computationally expensive, but it can be found in applications where uncertainty is large, such as electric vehicle charging under variable grid conditions [8]. Hybrid methods make use of offline initialisation coupled with online refinement, with a tradeoff between computational speed and flexibility [2]. One such critical gap in the literature is that the selection of a particular paradigm to be applied in a given application is not systematically advised.

## III. Taxonomic Framework: Controllers, Metaheuristics, and Objective Functions

Across the reviewed corpus, the practical value of tuning depends on interactions among three tightly coupled elements:

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controller architecture, search mechanism, and objective function. These dimensions jointly determine search space dimensionality, the physical meaning of the optimization, and the credibility of reported improvements. This section develops a three-layer taxonomic framework and interprets the literature through interactions rather than isolated algorithm labels.

### A. Controller Families

As shown in Figure 2, the reviewed corpus exhibits substantial controller diversity, though proportional–integral–derivative (PID) designs remain dominant. Classical integer-order controllers, including PI, PID, two-degree-of-freedom (2-DOF) PID, cascaded or multi-loop PI structures, and mixed PID+PI arrangements, appear in bidirectional electric vehicle (EV) fast charging [6], automatic generation control (AGC) [1], rehabilitation robotics [5], and photovoltaic (PV)-powered EV charging [9]. In these studies, the controller architecture is conventional, but tuning becomes nontrivial due to coupling, disturbance sensitivity, or multi-loop coordination.

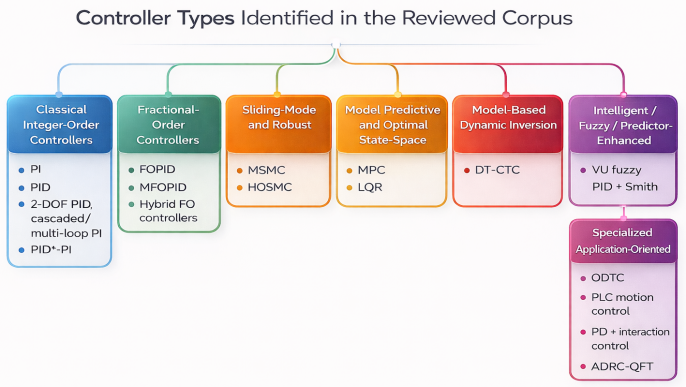



Fig. 2. Taxonomy of controller families in the reviewed metaheuristic-tuning corpus.

A second class comprises fractional-order and hybrid fractional-order controllers, where tuning extends beyond gain selection to include fractional orders. This group includes quadrotor fractional-order PID (FOPID) design [3], modified FOPID (MFOPID) exoskeleton control [7], and hybrid fractional-order AGC regulators [10]. The search problem is structurally richer because optimizers act on higher-dimensional, more strongly coupled parameter sets.

A third group includes sliding-mode and robust nonlinear controllers. Representative examples include modified exponential reaching law-based sliding-mode control for quadcopter tracking [11], high-order sliding-mode control (HOSMC) for vertical takeoff and landing (VTOL) trajectory tracking [4], and active disturbance rejection control combined with quantitative feedback theory (ADRC-QFT) for distillation control [12]. In these studies, optimization directly addresses disturbance rejection, robustness, and nonlinear closed-loop shaping.

The corpus also contains model-based optimal controllers, particularly model predictive control (MPC) and linear quadratic regulator (LQR). In a DC microgrid study, the optimizer tunes MPC weights [13]; in a three-degree-of-freedom (3-DOF) hover system study, it tunes LQR weighting matrices [14]. Here, optimization shifts from direct gain tuning to internal cost-structure design.

### B. Metaheuristic Optimizer Families

Figure 3 shows that swarm intelligence methods constitute the most frequent optimizer family. Particle swarm optimization (PSO) appears in quadcopter tracking [11], sliding-mode unmanned aerial vehicle (UAV) tuning [7], mobile manipulator control [15], rehabilitation robotics [5], and as part of a continuous stirred-tank reactor (CSTR) tuning framework [2]. Artificial bee colony (ABC) is used in EV fast charging [6], rehabilitation robot benchmarking [5], and coordinated PV-assisted fast-charger retuning [8]. Grey wolf optimizer (GWO) appears in vehicle-to-grid (V2G) converter tuning [16], while multi-objective GWO (MOGWO) is employed in hover-system LQR design [14] and ADRC-QFT distillation control [12]. This prevalence suggests that swarm-based methods remain attractive due to flexibility and applicability to nonlinear control design.

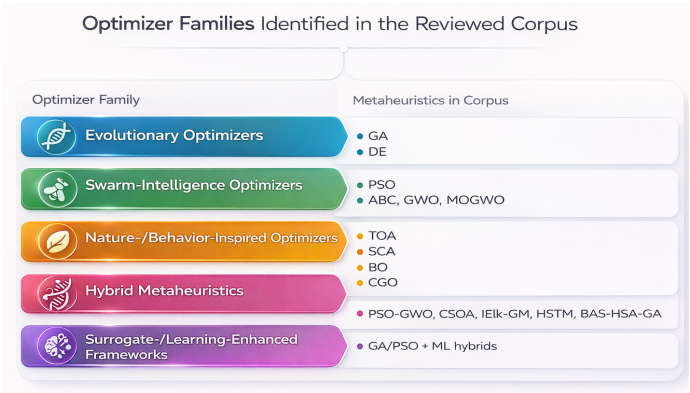



Fig. 3. Classification of optimizer families across the reviewed metaheuristic-tuning corpus.

Evolutionary optimizers form the second major family. Genetic algorithms (GA) are used in AGC [1], VTOL HOSMC tuning [4], industrial electric drive optimization [17], and within comparative or hybrid frameworks [2], [18]. Differential evolution (DE) appears in robot-environment interaction control [19]. These methods are particularly visible where broad search capability, benchmark comparison, or robust tuning is emphasized.

A third family includes nature-inspired optimizers beyond standard swarm/evolutionary groups, including tunicate optimization algorithm (TOA) in direct torque control (DTC)-based EV drive study [20] and Bonobo optimization (BO) in robust quadrotor fractional-order control [3]. Their presence reflects continued interest in algorithmic novelty, though methodological value depends on comparison and validation rigor.

Hybrid metaheuristics occupy a distinct position. The IElk-GM algorithm is used in MFOPID exoskeleton design [7];

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chaotic sunflower optimization algorithm (CSOA) appears in restructured AGC [10]; hybrid sine tree mean (HSTM) is introduced in PV-powered EV charging [9]; and beetle antennae search-harmony search algorithm-GA (BAS-HSA-GA) is used for unmanned surface vehicle (USV) heading control [18]. In these studies, the optimizer itself becomes part of the claimed contribution, as performance gains are linked to exploration–exploitation balance.

Finally, one study introduces a surrogate- or learning-enhanced framework, combining GA, PSO, and machine learning for robust PID tuning in a CSTR [2]. Although limited in the current corpus, this category is methodologically important as it indicates a transition from standalone search toward optimization-assisted intelligent design.

### C. Objective Function Families

As summarized in Table I, the reviewed corpus is not governed by a single performance logic; instead, six recurrent objective function families can be identified. The first family comprises classical error-integral formulations: integral of squared error (ISE), integral of absolute error (IAE), integral of error (IE), and mean squared error (MSE). These appear in bidirectional EV fast charging [6], rehabilitation robotics [5], CSTR control [2], and active PV-EV converter design [9]. In these cases, optimization is framed primarily as direct error minimization.

The second family comprises time-weighted error criteria: integral of time-weighted absolute error (ITAE), integral of time-weighted squared error (ITSE), and integral of squared time-weighted squared error (ISTSE). These appear in AGC under load variations [1], smart-grid frequency management [21], exoskeleton MFOPID tuning [7], restructured AGC [10], hover-system LQR design [14], and USV heading control [18]. These criteria penalize persistent deviations more strongly and are therefore especially relevant where long-tail error matters.

The third family comprises composite time-domain cost, combining error with settling time, control effort, smoothness, or related transient indices. This formulation appears in EV drive control [20], quadcopter sliding-mode control [11], VTOL HOSMC [4], mobile manipulator control [15], V2G converter control [16], PV-assisted fast-charger retuning [8], and mobile robot interaction control [19]. These formulations are methodologically stronger because they capture engineering trade-offs more realistically than single-index error criteria.

The fourth family comprises energy- or physics-aware objective design. The EV drive optimal direct torque control (ODTC) study explicitly targets torque ripple, losses, efficiency, and total harmonic distortion (THD) [20]; the industrial electric drive study optimizes energy consumption, efficiency, and losses [17]; and the mobile manipulator study includes energy-related terms [15]. These formulations connect tuning directly to physically meaningful system behavior.

The fifth family comprises structured predictive or optimal cost. In a DC microgrid study, the optimizer tunes internal MPC weights [13]. In the hover-system study, it tunes LQR $Q$ and $R$ matrices [14]. In both cases, optimization operates on the controller's internal decision criterion rather than on direct closed-loop gains.

The sixth family comprises robust or frequency-domain objective design. The CSTR study incorporates robustness-oriented min–max logic [2]. The restructured AGC study uses penalty-based terms under delay and energy storage system (ESS) conditions [10]; and the ADRC-QFT distillation study formulates tuning as a frequency-domain multi-objective optimization problem [12]. These studies embed uncertainty and robustness directly into the optimization task.

The reviewed studies reveal a tight coupling between controller type, optimizer family, and objective function design. Simple PI/PID structures typically pair with swarm or evolutionary methods and compact cost functions, where the main challenge is tuning under coupling or disturbances. Higher-dimensional controllers (e.g., fractional-order, HOSMC, fuzzy-PID) demand more complex search spaces, making the objective function increasingly influential in governing robustness-smoothness-tracking trade-offs, so improvement arises from controller–optimizer co-design rather than the optimizer alone. In MPC, LQR, and ADRC-QFT frameworks, the optimizer's role becomes structurally more important as it shapes internal cost weights or frequency-domain constraints. Finally, classical error integrals suffice for basic regulation, but applications involving ripple, losses, THD, or energy efficiency require application-relevant physical criteria to yield analytically meaningful results.

## IV. Application-Oriented Synthesis

Metaheuristic-based controller tuning has expanded across a broad range of engineering systems, but its practical significance is not uniform. As shown in Table II, the contribution of metaheuristics depends on plant nature, controller structure, performance objective, and operational constraints. This application-oriented perspective reveals not only where metaheuristics have been used, but why they become particularly valuable in specific control settings.

In **EV charging, power converters, and electric drives**, the dominant control architecture is hierarchical cascaded PI supplemented with feedforward or MPPT logic. Metaheuristics—most frequently ABC, PSO, and GA—are deployed not for individual loop tuning but for joint system-wide coordination across coupled stages such as DC-link stabilization, grid synchronization, and battery current regulation. Objectives consistently extend beyond classical error minimization to include THD, power factor, torque ripple, and energy efficiency, reflecting the industrial requirement for power quality and loss reduction. The primary challenge here is cross-stage coupling and transient power reversal, making metaheuristics valuable for resolving interactions that defy sequential loop-by-loop tuning.

**Power-system frequency and energy management** applications exhibit a different structure. Controllers range from classical PID to cascaded PIDn-PI and linear MPC, with metaheuristics performing either direct gain optimization or—in the case of MPC—internal weight tuning. The objective

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TABLE I
OBJECTIVE FUNCTION FAMILIES IN THE REVIEWED CORPUS

| Family | Representative Forms | Optimization Focus | Representative Refs. |
|---|---|---|---|
| Classical error-integral | ISE, IAE, IE, MSE | Direct error minimization | [6], [5], [2], [9] |
| Time-weighted error | ITAE, ITSE, ISTSE | Penalize persistent error | [1], [21], [3], [7], [10], [14], [18] |
| Composite time-domain | error + settling time; error + effort; error + smoothness | Multi-criterion transient shaping | [20], [11], [4], [15], [16], [8], [19] |
| Energy-/physics-aware | torque ripple, losses, efficiency, energy consumption | Physical performance optimization | [20], [17], [15] |
| Structured predictive/optimal | MPC multi-term cost; Pareto-based weighting | Internal cost-structure tuning | [13], [14] |
| Robust/frequency-domain | min–max; penalty-based robust cost; frequency-domain objectives | Robustness-oriented design | [2], [10], [12] |

landscape emphasizes frequency restoration, tie-line regulation, and robustness under renewable variability, communication delays, and denial-of-service conditions. Notably, several studies explicitly benchmark multiple optimizers, confirming that metaheuristic selection materially influences performance under uncertainty. The high dimensionality of weight spaces in MPC formulations represents a particularly rich arena for metaheuristic application.

The **aerial, robotic, and motion-control** category is the most structurally diverse, spanning UAV tracking, quadrotor control, VTOL, mobile manipulators, exoskeletons, and omnidirectional robots. Here, controller complexity escalates to nonlinear and fractional-order structures (FOPID, HOSMC, SMC, MFOPID, LQR), and the optimization search space becomes correspondingly high-dimensional. Metaheuristics are essential for navigating non-convex landscapes where analytical tuning fails. A striking pattern is the predominance of time-domain performance indices (ISE, IAE, ITAE, ITSE) as cost functions, often weighted or composited to balance tracking, smoothness, and effort. The recurring comparison of multiple metaheuristics within single studies suggests that no universal optimizer dominates; rather, problem-specific characteristics guide selection.

In **industrial and process control**, metaheuristics extend into robust and frequency-domain formulations. The CSTR study introduces surrogate-assisted optimization to manage expensive function evaluations, while the distillation work embeds metaheuristic tuning within an ADRC-QFT framework, demonstrating integration with classical robust control theory. These examples point toward a future where metaheuristics complement rather than replace analytical design methods.

Finally, **marine control systems**—exemplified by USV heading regulation—introduce predictor-enhanced structures (Smith predictor with fuzzy PID) to compensate for significant delays and environmental uncertainty. Metaheuristics here must handle time-delay compensation and wave disturbance rejection, problem features less prominent in other domains. Synthesizing across domains, three conclusions emerge. First, metaheuristic value is highest when controller architectures involve coupling, nonlinearity, or high-dimensional parameter spaces that defeat sequential or analytical tuning. Second, objective function design must move beyond generic error integrals to incorporate domain-relevant physical criteria (THD, torque ripple, SoC regulation, energy efficiency) for analytically meaningful results. Third, the choice of metaheuristic matters demonstrably, yet no single algorithm uniformly outperforms others; benchmarking against multiple optimizers should become standard practice. These patterns inform the systematic guidance for paradigm selection identified as a gap in the introduction.

## V. ANALYSIS OF METHODOLOGICAL DIMENSIONS

Table III synthesizes the reviewed studies along key methodological axes: controller architecture, tuning paradigm, problem dimensionality, optimizer selection, validation rigor, and reported performance improvement. Rather than serving as a mere catalog, this comparison enables critical analysis of what makes metaheuristic tuning effective, where it adds genuine value, and where current literature exhibits gaps. The table reveals that controller complexity spans from classical low-order structures (PI, PID, PD) to advanced formulations (fractional-order, sliding-mode, MPC, ADRC-QFT, fuzzy Smith predictors). A clear pattern emerges: studies employing simple controllers often attribute performance gains primarily to optimizer choice, whereas studies with sophisticated controllers demonstrate co-design effects where controller structure and tuning jointly determine outcomes. For instance, [11] and [7] show that controller redesign combined with optimization yields improvements not achievable by tuning alone. Conversely, [16] demonstrates that even a simple two-parameter PI can perform strongly if systematically tuned, challenging the assumption that controller complexity must escalate to achieve high performance. The tuning paradigm across all reviewed studies is exclusively offline optimization. No study implements true online adaptation or hybrid offline–online refinement, confirming the gap identified in the introduction. This absence suggests that real-time metaheuristic tuning remains either computationally prohibitive or methodologically underdeveloped for most applications.

Several studies explicitly benchmark multiple metaheuristics, yielding actionable insights. A comparison of seven optimizers under renewable variability, delay, and denial-of-

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TABLE II
APPLICATION-ORIENTED CLASSIFICATION OF THE REVIEWED STUDIES.

| Ref. | Year | Domain | Objectives | Controller | Metaheuristic | Cost | Constraints |
|---|---|---|---|---|---|---|---|
| **EV Charging, Power Converters, and Electric Drives** | | | | | | | |
| [6] | 2025 | EV fast chg. | DC-link stability; G2V/V2G coord.; batt. curr. ctrl.; low THD; near-unity PF | Hierarchical cascaded PI + batt.-curr. FF | Joint system-wide PI tuning via ABC | Wtd. ISE + ITAE | Cross-stage DC-link coupling; THD compliance; batt.-curr. ripple; transient power-flow reversal |
| [20] | 2025 | EV traction drives | Torque-ripple reduction; loss minimization; eff. improvement; lower THD | ODTC w/ dynamic ref.-flux selection | TOA-based ref.-flux optimization | Wtd. multi-obj. | Motor-param. sensitivity; admissible flux bounds; torque-capability constraint |
| [17] | 2026 | VC electric drives | $(E_{con})$; EE; losses | PLC-based motion ctrl. | GA motion-profile tuning | Single-obj. | Time budget; VC/DT fidelity |
| [8] | 2026 | PV-assisted EV fast chg. | PV; DC-link; THD; PLL | Coordinated cascaded PI + MPPT | ABC retuning with PSO comparison | ISE + ITAE | PV–grid coupling; THD; transitions |
| [8] | 2026 | PV-assisted EV fast chg. | PV; DC-link; THD; PLL | Coordinated cascaded PI + MPPT | ABC-PSO re-tuning | ISE + ITAE | PV–grid coupling; THD; transitions |
| [9] | 2025 | PV EV charging | Gain; regulation; fast resp. | HSTM-PI | STO–MOA hybrid tuning | Volt.-err.-based cost | PV nonlinearity; stress; EMI |
| **Power-System Frequency and Energy Management** | | | | | | | |
| [1] | 2025 | AGC/LFC | Freq. restoration; tie-line regulation; OS/US reduction under wide LV cases | GA-PID | GA-based PID tuning | ITAE | 100–300 MW load steps; inter-area coupling; dynamic LV scenarios |
| [21] | 2026 | Smart-grid AGC | $(\Delta F_1)$, $(\Delta F_2)$, $(\Delta P_{tie})$ minimization; robust. under RES/delay/DoS | Cascaded PIDn-PI | Gain tuning + multi-MH benchmark | ITAE | RES variability; comm. delay; DoS; multi-source nonlinear dynamics |
| [13] | 2025 | DC microgrid MPC | Tracking; SoC regulation; smooth dispatch; control-effort balancing | Linear MPC | MPC weight tuning via PSO/GA/Pareto/Pattern Search | Multi-term weighted MPC cost + tracking-loss/regularization | High-dim. weight space; parameter interdependency; forecast/model reliance |
| [10] | 2025 | Restructured AGC/LFC | $(\Delta F)$; $(\Delta P_{tie})$; robust. | Hybrid FO ctrls. | CSOA tuning | ISTSE + penalty | Market cases; RLD; CTD; ESS coord. |
| **Aerial, Robotic, and Motion-Control Systems** | | | | | | | |
| [11] | 2025 | UAV tracking | Pos./att.; robust.; anti-chat. | Modified ERL-SMC | PSO tuning | IMSE/MSE | Nonlinearities; disturb.; uncert.; sim.-only |
| [3] | 2025 | Quadrotor FO ctrl. | OS; $(t_r)$; $(t_s)$; robust. | PID/FOPID | BO tuning + benchmark | Time-domain perf. index | High-dim. tuning; comp. burden |
| [4] | 2025 | VTOL tracking | 3D traj.; smoothness; eff. | HOSMC | GA gain opt. | Wtd. composite | Turbulence; perturb.; transition dynamics |
| [15] | 2026 | Mobile manipulators | Tracking; energy; real-time | DT whole-body CTC | PSO tuning | Wtd. RMS err. + energy | Coupling; ZOH; nonholonomic; embedded limits |
| [5] | 2025 | Rehab robotics | Smooth motion; low OS/$(t_s)$ | PID / 2-DOF PID | PSO/FF/ACO/ABC benchmark | ISE / IAE | Safety; disturb.; limited RT |
| [7] | 2025 | Lower-limb exoskel. | Tracking; low SSE/OS; smooth torque | MFOPID | IElk-GM tuning | ITAE | Nonlinear dyn.; uncert.; torque bounds |
| [14] | 2026 | 3-DOF hover UAV | Roll/pitch/yaw tracking | LQR | MOGWO $(Q, R)$ tuning | IAE / ITAE / ISE / ITSE | Coupled axes; saturation; sim.-only |
| [19] | 2025 | Omni mobile robots | Tracking; smoothness; obstacle resp. | PD + interaction ctrl. | DE tuning | ISE + ISDU | Obstacles; sensing limits; actuator bounds |
| **Industrial and Process-Control Systems** | | | | | | | |
| [2] | 2025 | CSTR control | Tracking; robust.; disturb. rej. | Gain-scheduled PID | GA/PSO + ML surrogate | ISE / min–max | Nonlinearity; worst-case plants; ML need |
| [12] | 2025 | Distillation control | Tracking; decoupling; robust. | ADRC-QFT | Improved MOGWO tuning | Freq.-domain multi-obj. | Coupling; uncert.; spec. conflict |
| **Marine Control Systems** | | | | | | | |
| [18] | 2025 | USV heading ctrl. | Heading; delay comp.; robust. | VU fuzzy PID + Smith | BAS-HSA-GA tuning | Time-wtd. sq. err. | Delay; waves; marine uncert. |

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TABLE III
METHODOLOGICAL COMPARISON OF CONTROLLER TUNING STUDIES

| Ref. | Year | Controller Type | Paradigm | Params. | MH(s) | Key Finding | Val. | Methodological Insight |
|---|---|---|---|---|---|---|---|---|
| **EV Charging, Power Converters, and Electric Drives** | | | | | | | | |
| [6] | 2025 | Cascaded PI + feedforward | Classical coordinated multi-loop control | 6 | ABC | Weak transients → 920→238 ms, US removed, THD $< 1.2\%$ | Sim. | Sys.-lvl co-design; gains from coord. cross-stage tuning, not iso. loop tuning |
| [20] | 2025 | ODTC with optimal flux reference | Optimization-enhanced direct torque control | 1 | TOA | Fixed-flux losses/ripple → eff. ↑, ripple/THD/energy ↓ | Sim. | Perf. strongly model-dependent; essentially flux-opt.-driven ctrl. |
| [17] | 2026 | PLC motion control | Industrial motion-profile optimization | 3 | GA | Exhaustive-search ref. → GA matched optimum; up to 114× faster | VC + HIL | Highly practical + rigorously val.; yet low-dim. problem eases tuning burden |
| [16] | 2025 | Cascaded SRF-PI | Classical converter control | 2 | GWO | Manual PI/ANFIS weaker → GWO-PI best | Sim. + HW | Even simple PI can perform strongly if systematically tuned |
| [8] | 2026 | Multi-loop PI + MPPT | Coordinated multi-converter control | 6 | ABC | MPPT-alone weak → THD 4.02→1.40%, $(t_s)$ 442→253.8 ms | Sim. | MPPT alone insufficient; full-system retuning required post-int. |
| [9] | 2025 | PI | Classical voltage control | 2 | HSTM | GA/PSO-PI weaker → HSTM-PI best | Sim. + HW | Promising result, but incomplete optimizer details weaken reprod. |
| **Power System Frequency and Energy Management** | | | | | | | | |
| [1] | 2025 | Three PID controllers | Classical AGC/LFC | 9 | GA | CAGC/TLB weaker → OS ↓ up to 90%, $(t_s)$ ↓ 47% | Sim. | Broad LV eval.; relevance ↑, but val. limited to sim. |
| [21] | 2026 | PIDn-PI cascaded controller | Classical cascaded AGC with metaheuristic benchmarking | 12 | SCA, GWO, PSO-GWO, TSA, SSA, PSO, CGO | Slower/poorer MHs → SCA best | Sim. | Mainly benchmarking; contrib. centers on MH comparison > ctrl. novelty |
| [13] | 2025 | Linear MPC | Predictive control | 20 | PSO, GA, Pareto Search, Pattern Search | GA ∼16% err. → PSO ∼1.9% | Sim. | Offline weight tuning; useful for MPC cost design, but optimizer-/system-spec. |
| [10] | 2025 | Hybrid fractional-order controllers | Hybrid robust/intelligent AGC control | 6–8 per area | CSOA | Weaker optimized ctrls. → CSOA hybrids best | Sim. | Hybrid ctrl. arch. + ESS int. jointly shape AGC perf. |
| **Aerial, Robotic, and Motion-Control Systems** | | | | | | | | |
| [11] | 2025 | Modified sliding mode control (MSMC) | Robust nonlinear control | 18 | PSO | SMC variants weaker/chattering → MSMC best | Sim. | Gains stem from ctrl. redesign + tuning; not attributable to opt. alone |
| [3] | 2025 | PID / FOPID | Classical vs fractional-order control | 12 / 20 | GA, PSO, GWO, BO | PID weaker → BO-FOPID best, 0% OS | Sim. | FOPID clearly dominates PID here; MH comparison is secondary |
| [4] | 2025 | HOSMC | Robust nonlinear sliding-mode control | 4 | GA | Alt. ctrls. weaker → HOSMC best | Sim. | Strong comparative baseline; evidence still sim.-only |
| [15] | 2026 | Discrete-time computed torque control (DT-CTC) | Model-based nonlinear control | Gain vector | PSO | DT-PD weaker → DT-CTC better in sim./lab | Sim. + HW | Impl.-oriented study w/ real HW; comparison mainly vs PD |
| [5] | 2025 | 2-DOF PID | Classical control with metaheuristic tuning | 6 | PSO, Firefly, ACO, ABC | ZN poor → MHs much better; ABC best overall | HW | Good practical evidence; ranking is metric-dependent (ISE vs IAE) |
| [7] | 2025 | FOPID / MFOPID | Fractional-order control | 14 / 18 | IElk-GM | FOPID slower → MFOPID: 0% OS, lower ITAE | Sim. | Gain reflects struct. modif. + opt.; clear co-design effect |
| [14] | 2026 | LQR | Optimal state-feedback control | 10 | MOGWO | — → Pareto-selected best compromise | Sim. | Mainly a weight-selection framework; contrib. in multi-obj. tuning, not new ctrl. |
| [19] | 2025 | PD + interaction controller | Robot–environment interaction control | 6 | DE | CCTA weaker → $(J)$ ↓ 61.57% in sim. | Sim. + HW | Strong val. via sim., expt., + force-sensor confirmation |
| **Industrial and Process Control Systems** | | | | | | | | |
| [2] | 2025 | Gain-scheduled PID | Adaptive / robust PID with ML enhancement | 3+ | GA, PSO, ML-enhanced hybrids | GA/PSO limited → ML hybrids fastest/most robust | Sim. | Adversarial robust tuning is methodol. strong; reporting uneven, no HW val. |
| [12] | 2025 | ADRC-QFT | Robust disturbance-rejection control | ∼4 per loop | MOGWO | PI-QFT/MPC weaker → ADRC-QFT best | Sim. | QFT-guided opt. is methodol. distinctive; adds rigorous freq.-domain basis |
| **Marine Control Systems** | | | | | | | | |
| [18] | 2025 | Enhanced Smith predictor + variable-universe fuzzy PID | Hybrid delay-compensated intelligent control | 5 | BAS-HSA-GA | PID weaker → $(t_s)$ 37.8 s, OS 0.3% under 2 s delay | Field sim. | Among strongest val. levels; notable for explicit delay-robustness testing |

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service conditions finds that the Sine Cosine Algorithm (SCA) outperforms more frequently used algorithms like PSO and GWO [21]. Bayesian optimization (BO) tuned fractional-order PID achieves 0% overshoot, outperforming GA, PSO, and GWO—suggesting that BO's probabilistic surrogate model suits expensive, non-convex evaluations [3]. Optimizer ranking depends on the chosen cost metric (ISE vs. IAE), a critical nuance often overlooked [5]. Collectively, these findings indicate that no universal optimizer dominates; problem characteristics and objective formulation materially influence selection.

The number of tuned parameters varies dramatically: from single-parameter flux optimization to 20-dimensional MPC weight tuning [13], [20]. Higher-dimensional problems do not automatically justify metaheuristic use—low-dimensional problems (three parameters) benefit primarily from computational speedup (114× faster than exhaustive search), not global search capability [17]. Conversely, metaheuristics add value through coordinated cross-stage tuning in multi-loop systems, where interactions defeat sequential loop-by-loop methods [6], [8]. Objective functions also vary from single-term error integrals to weighted composites incorporating THD, torque ripple, energy efficiency, and SoC regulation. Studies using application-relevant physical criteria yield more analytically meaningful results than those relying solely on generic ISE/ITAE minimization [6], [20].

Validation approaches span four levels: simulation only, simulation with hardware-in-the-loop (HIL), simulation plus laboratory hardware, and field validation. Only one study includes field validation, representing the gold standard [18]. Several studies provide credible hardware validation, confirming that simulation findings translate to physical systems [15], [16], [19]. However, a substantial portion of the reviewed corpus, including otherwise methodologically strong studies, remains simulation-only, limiting claims of practical relevance [3], [4], [12]. The table also reveals reporting deficiencies: one study presents promising results but provides incomplete optimizer details, undermining reproducibility [9].

Quantified performance improvements vary widely, from modest (error reduction from 16% to 1.9%) to dramatic (settling time reduction from 920 ms to 238 ms) [6], [13]. However, careful attribution is often lacking. Many studies treat performance gains as attributable to the optimizer alone, ignoring the confounding effect of controller structure changes, objective function design, or baseline selection. A few studies are exceptions, explicitly discussing co-design effects [7], [11]. A recurring weakness is comparison against poorly tuned or outdated baselines (e.g., Ziegler–Nichols), which inflates perceived metaheuristic advantage.

## VI. Research Gaps and Future Directions

Our synthesis reveals five critical gaps constraining the field's progress. First, objective function design remains disconnected from engineering priorities. While most studies employ generic error integrals (ISE, ITAE), these criteria neglect ripple, efficiency, THD, and actuator wear. The most convincing studies incorporate application-relevant physical criteria [8], [17], [20], yet such formulations remain exceptional. Second, constraint handling is systematically underdeveloped. Most studies treat constraints as penalty terms rather than inviolable bounds, leading to parameters that may violate saturation, thermal limits, or safety margins. Exceptions include the min–max robustness formulation [2] and penalty-based constraints [10], but standardized approaches remain absent. Third, benchmarking fairness is problematic. Comparisons against classical methods or alternative metaheuristics rarely standardize computational budget, population size, iteration count, or hyperparameter tuning. Rigorous comparisons like those using consistent benchmarks across seven optimizers [21] or fair comparisons of GA, PSO, GWO, and BO [3] remain uncommon. Fourth, reproducibility is severely limited. Few studies report hyperparameter settings (population size, iteration count, mutation/crossover rates, inertia weights) or provide code repositories, preventing independent replication and cumulative progress. Fifth, validation depth varies dramatically. Hardware-in-the-loop or experimental validation appears in a minority of studies [5], [9], [15]–[19], and field validation under realistic conditions remains exceptionally rare.

Addressing these gaps requires actionable changes. For objective functions, future work should move toward cost functions reflecting engineering priorities: torque ripple and efficiency for drives, THD and power factor for converters, energy consumption for motion systems, and robustness margins for uncertain plants. Pareto-based multi-objective optimization should replace weighted sums when objectives conflict [14]. For constraint handling, hard constraints should be encoded through repair mechanisms, feasibility-preserving operators, or constrained domination principles. Standardized formulations for saturation, rate limits, and thermal bounds would enable fair cross-study comparison. For benchmarking, the field needs standardized testbeds with defined computational budgets, initialization protocols, and evaluation metrics. Studies should report hyperparameter settings in standardized tables and, where possible, provide code repositories. For validation, a tiered approach is recommended: simulation for feasibility and tuning, HIL for real-time validation, hardware for proof-of-concept, and field deployment for ultimate credibility. Each tier should be justified based on research questions and application requirements. Finally, the interaction among controller architecture, optimizer characteristics, and objective function design suggests that future progress will be driven by problem-structure-aware optimization rather than algorithm-centric competition. Researchers should ask not "which optimizer is best?" but "given my controller structure, plant dynamics, and performance priorities, what optimization approach is most appropriate?"

## VII. Conclusion

This review has provided a critical synthesis of metaheuristic-based controller tuning in engineering systems based on studies published between 2025 and 2026. Organized thematically around controller families, optimizer families,

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and objective function types across major application domains, three main findings emerge. First, while controller diversity has expanded from classical PID to fractional-order, sliding-mode, and MPC structures, most studies employ relatively simple controllers. Tuning challenges arise from coupling, disturbances, and multi-loop interaction rather than controller complexity alone. Second, methodological quality varies dramatically. The most convincing work is characterized by objective functions reflecting physical priorities, constraint handling respecting operational limits, fair benchmarking against meaningful baselines, and validation extending beyond simulation. Third, metaheuristic value is strongly application-dependent: most valuable for multiloop coordination in EV charging, coupling and uncertainty in power systems, nonlinearities and multi-criterion tracking in robotics, and delay compensation in marine systems.

The field faces persistent gaps in objective design, constraint handling, benchmarking fairness, reproducibility, and validation depth. Addressing these requires shifting focus from algorithmic novelty to methodological rigor. Four recommendations emerge from this synthesis. First, benchmarking against multiple optimizers with metric-dependent ranking should become standard practice. Second, validation beyond simulation—particularly hardware or field testing—is essential for practical applicability. Third, performance improvements must be attributed transparently, distinguishing optimizer contribution from controller redesign or objective formulation. Fourth, reporting standards must improve to include optimizer hyperparameters, convergence behavior, and computational cost.

We conclude that metaheuristic-based controller tuning has established itself as a valuable methodology, but its future impact depends less on new algorithms than on problem-structure-aware optimization frameworks that are physically meaningful, application-aware, and methodologically credible. The most important question for researchers is not "which optimizer is best?" but "how should the tuning problem be formulated to reflect real engineering priorities and constraints?"

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